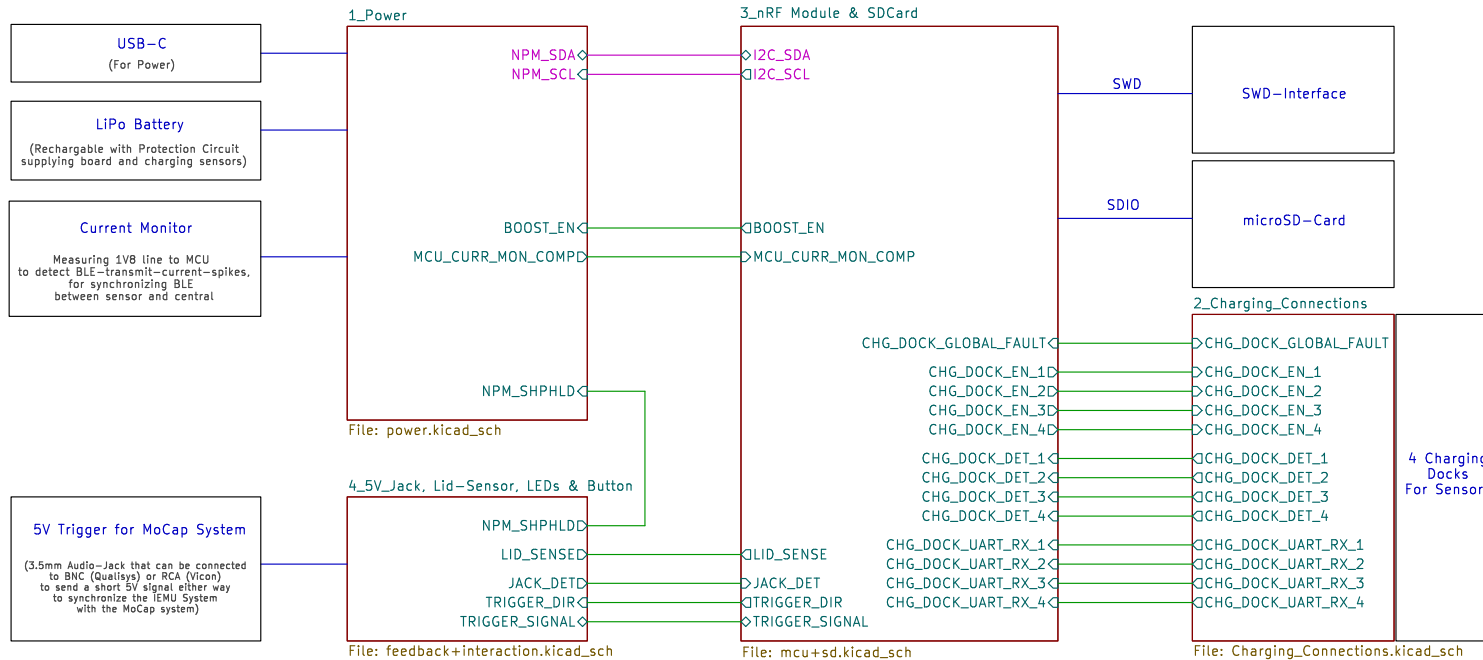


BASE – IEMU Development Board – Open–source Inertial Sensor for Biomechanics

Overview



Test Points

TP101	+VBUS_IN
TP102	+VBAT
TP103	–VBAT
TP104	+VSY5
TP105	+1V8_MCU
TP106	+3V0
TP107	+5V0
TP108	BOOST_EN
TP109	MCU_CURR_MON_COMP
TP110	GND
TP111	GND
TP112	GND
TP201	CHG_DOCK_EN_1
TP202	CHG_DOCK_5V_VOUT_1
TP203	CHG_DOCK_DET_1
TP204	CHG_DOCK_UART_RX_1
TP205	CHG_DOCK_EN_2
TP206	CHG_DOCK_5V_VOUT_2
TP207	CHG_DOCK_DET_2
TP208	CHG_DOCK_UART_RX_2
TP209	CHG_DOCK_EN_3
TP210	CHG_DOCK_5V_VOUT_3
TP211	CHG_DOCK_DET_3
TP212	CHG_DOCK_UART_RX_3
TP213	CHG_DOCK_EN_4
TP214	CHG_DOCK_5V_VOUT_4
TP215	CHG_DOCK_DET_4
TP216	CHG_DOCK_5V_VOUT_4
TP301	SWD_CLK
TP302	SWD_IO
TP303	SWD_RESET
TP304	SDIO_CMD
TP305	SDIO_CLK
TP306	SDIO_D0
TP307	GND
TP308	MUX_UART
TP309	I2C_SCL
TP310	I2C_SDA
TP401	TRIGGER_SIGNAL
TP402	TRIGGER_DIRECTION
TP403	TRIGGER_SIGNAL_AFTER_LS
TP404	NPM_SHPHLD_BUTTON

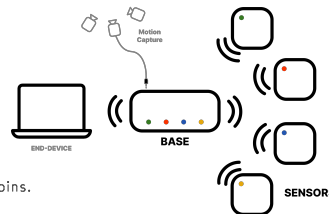
Concept

This is the development board for the base of 4 sensors, that will be used for biomechanical measurements of acceleration and angular velocity (IMU). The base acts as a charger for the sensors. In this iteration via cable, in the future via docks and pogo pins.

It also is the central coordinator between the user device and the sensors. All sensors send its data to the base and the base has more storage capacity of said data via its microSD storage.

Most importantly the base synchronizes the sensors via providing a source of truth timestamp. This timestamp is matched via observing the BLE current spike during transmit (current monitor) on both the base and sensor side.

The base also provides a 3.5mm Audio–Jack, that can send or receive a 5V Trigger from a Motion Capture System like Vicon or Qualisys, allowing synchronization between the sensors and these external systems.



BASE – IEMU Development Board – Open–source Inertial Sensor for Biomechanics
Mustafa Acar

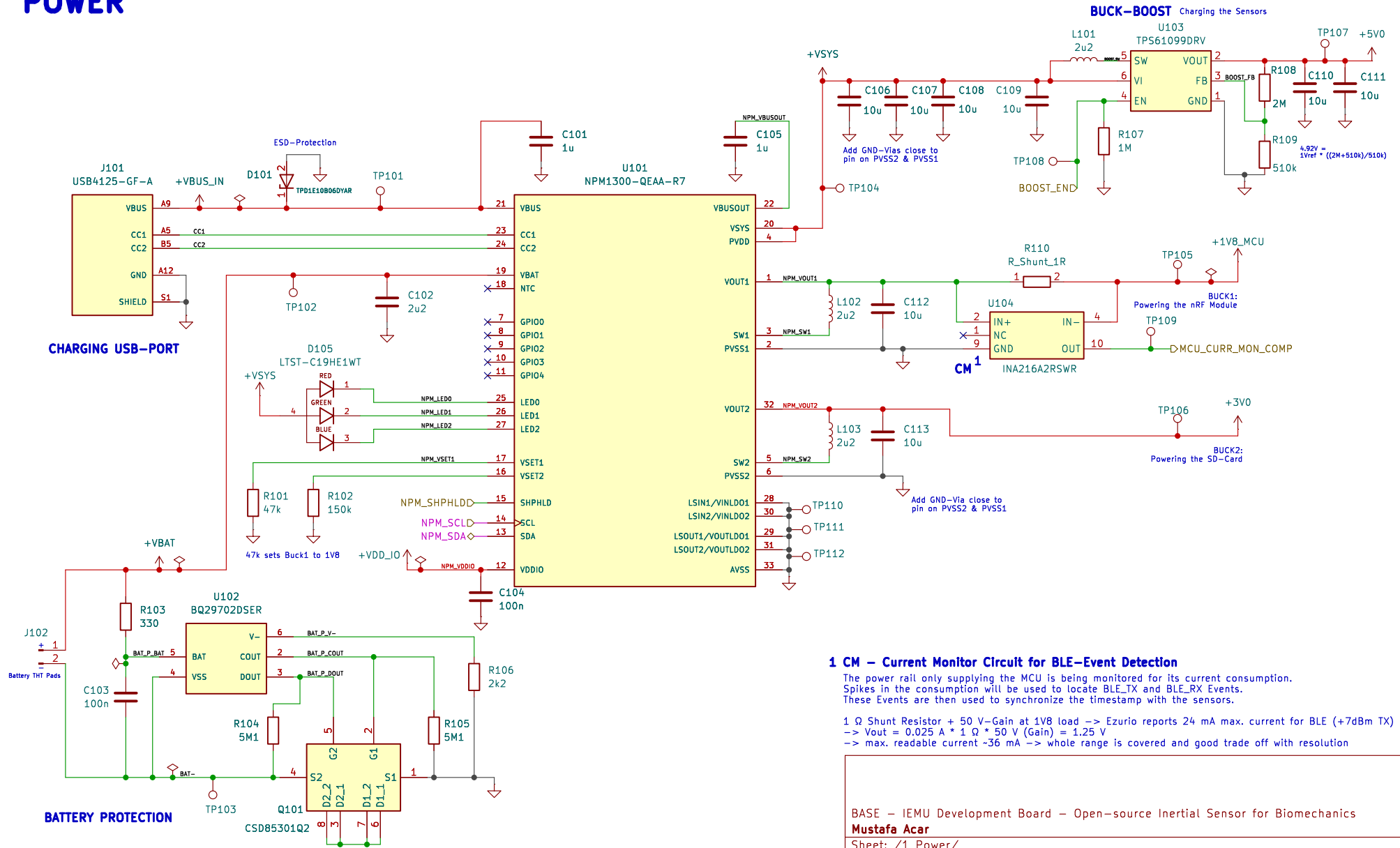
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Size: A4 Date: 2026–05–04
KiCad E.D.A. 9.0.5

Rev:
Id: 1/5

POWER



1 CM – Current Monitor Circuit for BLE–Event Detection

The power rail only supplying the MCU is being monitored for its current consumption. Spikes in the consumption will be used to locate BLE_TX and BLE_RX Events. These Events are then used to synchronize the timestamp with the sensors.

1 Ω Shunt Resistor + 50 V-Gain at 1V8 load -> Ezurio reports 24 mA max. current for BLE (+7dBm TX)
 -> $V_{out} = 0.025 \text{ A} * 1 \Omega * 50 \text{ V (Gain)} = 1.25 \text{ V}$
 -> max. readable current ~36 mA -> whole range is covered and good trade off with resolution

BASE – IEMU Development Board – Open-source Inertial Sensor for Biomechanics

Mustafa Acar

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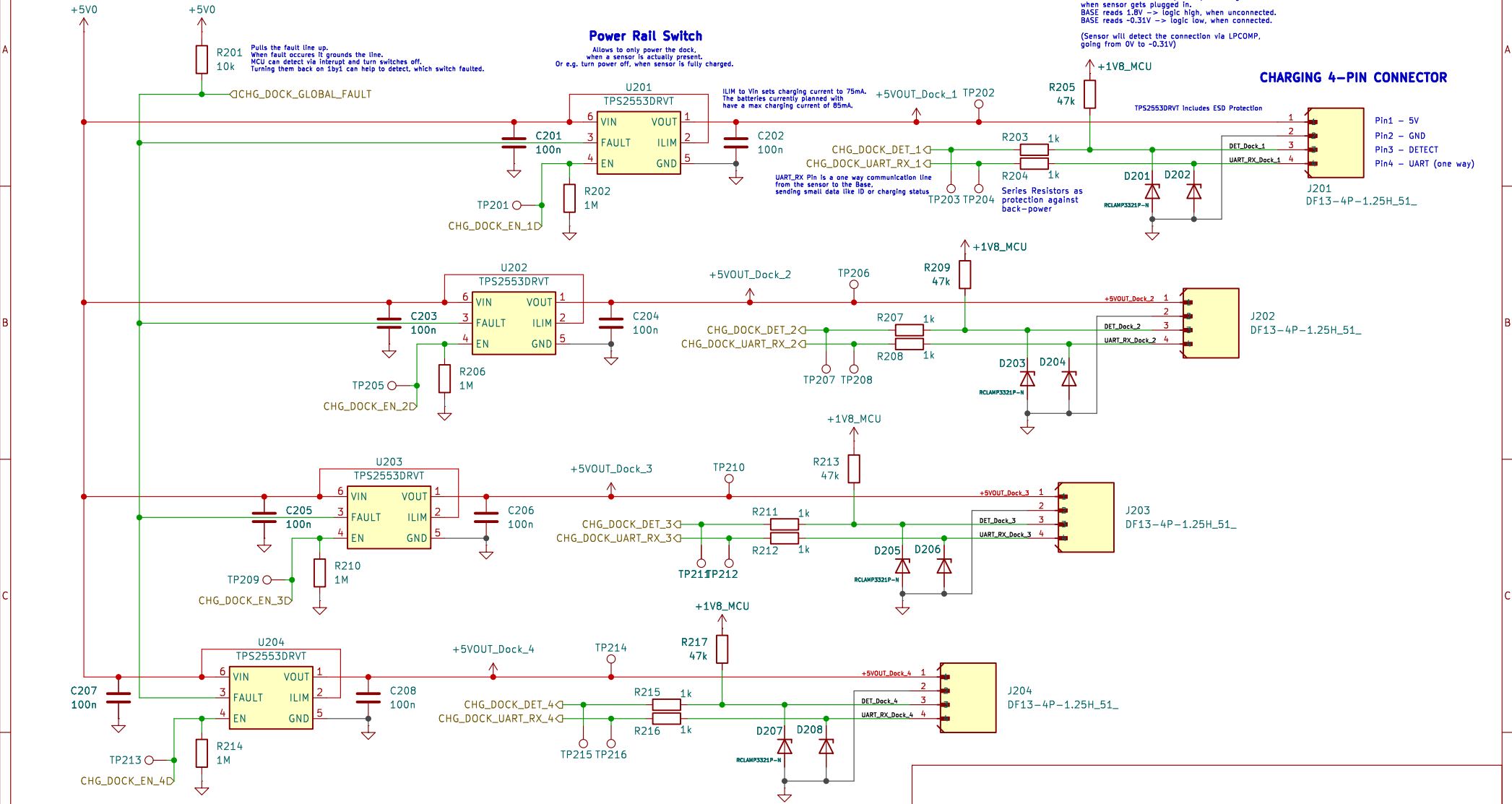
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Rev: 0.1

Id: 2/5

Charging Connections



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Mustafa Acar

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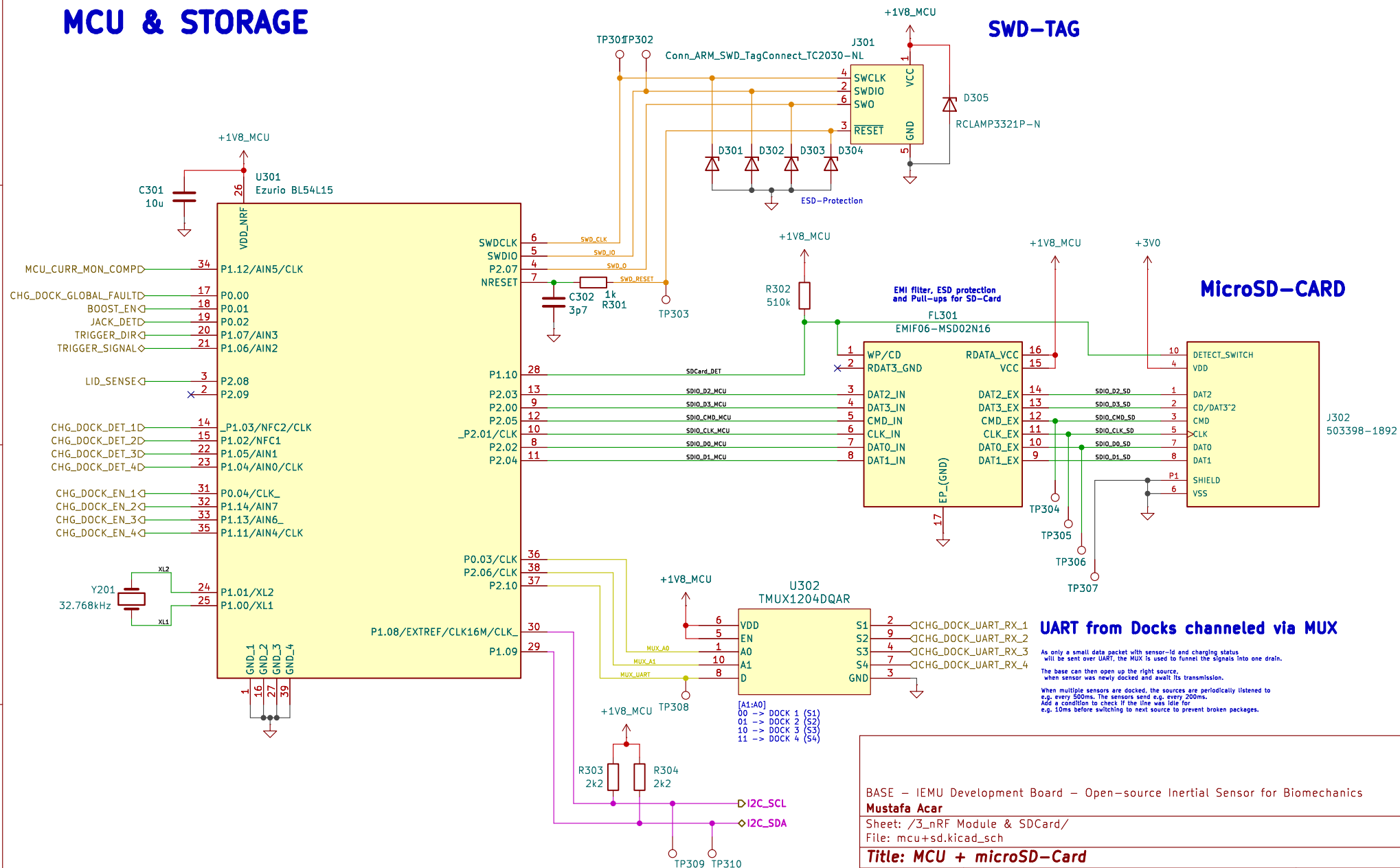
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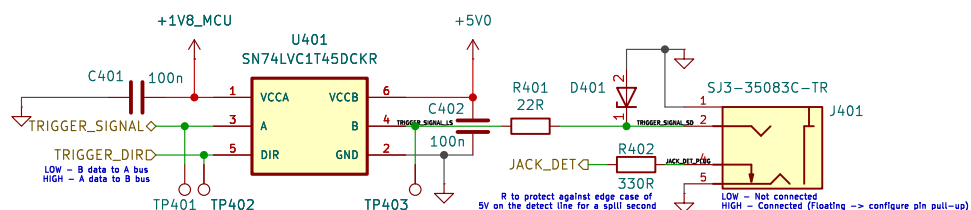
Rev: 0.1
Id: 3/5

MCU & STORAGE

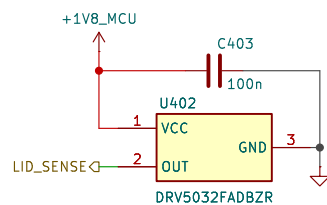


Connector, Lid-Sensor, LEDs & Buttons

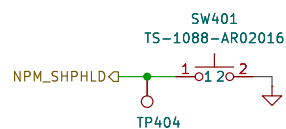
5V-Trigger-Jack



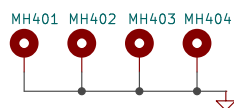
Lid-Hall-Effect-Sensor



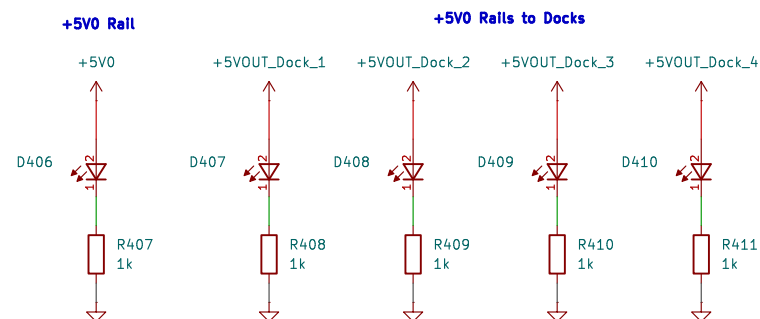
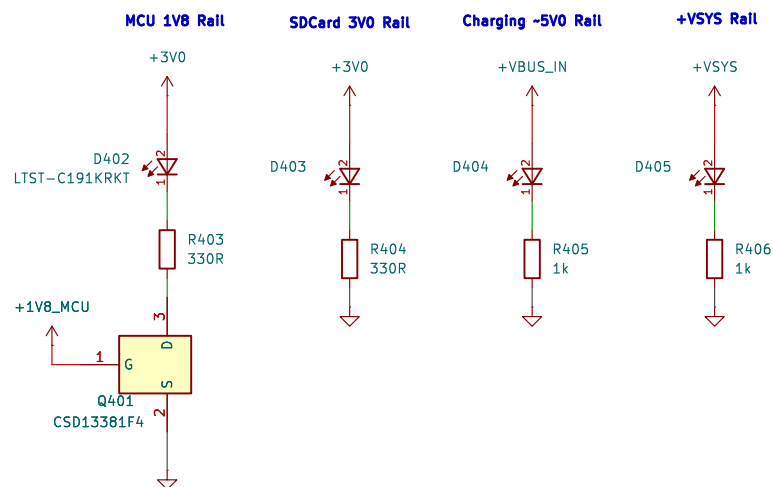
Buttons



Mounting Holes



Debug LEDs



BASE – IEMU Development Board – Open-source Inertial Sensor for Biomechanics
Mustafa Acar

Sheet: /4_5V_Jack, Lid-Sensor, LEDs & Button/
File: feedback+interaction.kicad_sch

Title: 5V-Jack, LEDs & Button

Size: A4	Date: 2026-05-07
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KiCad E.D.A. 9.0.5

Rev: 0.1

Id: 5/5